

Quantifying the effects of gene knockout at single cell resolution reveals lack of Stim2 affects vision-dependent behaviour via sensory neurons

Results

Quantification of the effects of Stim2 deletion at single cell resolution

We mapped the raw reads from the scRNA-seq experiments to the Zebrafish genome, quantified gene expression, removed potential cell doublets, integrated data from all samples, performed preprocessing, normalisation and log-transformation of the data, computed highly variable genes, computed the neighbourhood graph, subsequently embedded it using UMAP (Uniform Manifold Approximation and Projection) and performed Leiden clustering (see methods).

Although the common algorithms for community detection (clustering) on neighbourhood graphs provide high modularity in clustering, the lack of cell-attribute information causes the clusters to have lower purity³². Annotating such clusters based on marker genes highly expressed by only a certain percentage of member cells of the clusters could lead to erroneous/overgeneralized assignment of cell types. For this reason, we performed clustering of UMAP embedding of the neighbourhood (cell-cell similarity) graph with the Leiden algorithm, but did not force cell type annotations on the clusters. We subsequently utilised these clusters in a cell-type-agnostic way for quantifying the effects of stim2 deletion.

We identified a total of 30 clusters (0 through 29, Leiden clustering, resolution 1; Fig.1B) in the UMAP embedding of integrated transcriptomic data at the single cell resolution from neuronal origin cells from all the samples. PHATE (Potential of Heat-diffusion for Affinity-based Transition Embedding) embedding of the data is shown in Fig. 1C. We subsequently utilised the MELD algorithm³³ that describes the transcriptomic space as a manifold and utilises graph signal processing to calculate a relative likelihood estimate of observing each cell in each of the experimental conditions ($(stim2a;stim2b)^{-/-}$ and Tg ; see methods). We first estimated densities for each sample individually (Fig S***), and computed the sample associated relative likelihoods for each condition ($(stim2a;stim2b)^{-/-}$ and Tg ; Fig S1). Finally, the average likelihood of the four $(stim2a;stim2b)^{-/-}$ samples was used as the quantitative measure of stim2 deletion at single cell resolution (Fig. 1D). Each cell was assigned a likelihood score. The cells from the double-knockout zebrafish that were members of Cluster 5 had the highest mean $(stim2a;stim2b)^{-/-}$ likelihood (Fig 1E, farthest right; mean in grey dot) among all clusters, meaning these cells were the most affected by stim2 deletion. Also, a large majority (>62%) of the member cells in Cluster 5 were from $(stim2a;stim2b)^{-/-}$ zebrafish.

Gene set enrichment analysis revealed that the KEGG pathway - neuroactive ligand-receptor interaction - was significantly enriched ($P_{\text{adj}} = 1.284 \times 10^{-9}$) among the top differentially expressed (DE) genes of Cluster 5 (Bayes factor > 3.3, probability of differential expression > 0.9646, median log2 fold change > 1, Cluster 5 vs rest; 301 genes). The pathway is involved in environmental information processing and is composed of interacting signalling molecules. The top marker genes of Cluster 5 and their expression profile across all clusters are shown in Figure 2A. We then obtained cell type information of these marker genes from our curated list (see methods, Supplementary Table 1). A frequency-based visualisation of the cell type information (corresponding to the top genes selected with the criteria: non_zeros_proportion1 > 0 and Bayes Factor > 3.3; Supplementary Table 2) revealed the presence of an admixture of cell types in Cluster 5. Interneurons, dopaminergic neurons, CNS neurons, and hypocretin-secreting neurons were observed in highest frequencies as major classes of cells. Different subtypes of these cells (primary interneurons, CNS interneurons, amacrine cells, bipolar neurons from the class of interneurons; sensory processing neurons such as retinal ganglion cells, neural crest cells, chemoreceptor cells such as Kolmer-Agduhr neurons, olfactory granule cells and pyramidal cells from the class of CNS neurons, cranial nerve CN-X, efferent neuron such as secondary motor neuron etc) and other cells were present in the cluster (Fig 2B).

We then visualised (*stim2a;stim2b*)^{-/-} likelihood of all cells from each of the UMAP clusters on PHATE embedding (Fig 3A). As the UMAP clusters were sorted by increasing average likelihood (Fig 1E), UMAP Cluster 5 with the highest average likelihood is now shown at the last panel of Fig 3A (sorted cluster 29, 0-based indexing) and Fig 3B. Each of these clusters was further divided into 2-5 subclusters with vertex frequency clustering (VFC) (Supplementary

Figure S2) and the best subclustering level (between 2 and 5) for each cluster was selected based on optimal grouping of cells as observed in the distribution of likelihood values. Different levels of subclustering of the UMAP Cluster 5 is shown in Fig 3C (the level with three subclusters was chosen for downstream analysis). The (*stim2a;stim2b*)^{-/-} likelihood of cells from all the subclusters (at optimal level of grouping) from each of the selected 14 UMAP clusters (based on the number of member cells with high likelihood) is shown in Fig 3D. One of the subclusters from UMAP cluster 5 indeed had the highest average (*stim2a;stim2b*)^{-/-} likelihood (subcluster 63, farthest right, Fig 3D). The low-, mid-, and high-likelihood subclusters (4, 44, and 63, respectively) of the UMAP cluster 5, as seen on PHATE embedding (Fig 3E), showed high fidelity in segregating cells from *Tg* and (*stim2a;stim2b*)^{-/-} samples according to (*stim2a;stim2b*)^{-/-} likelihood (Fig 3F). Next, we identified the characterising genes for each subcluster with differential expression analysis (each subcluster vs rest of the cells). Differentially expressed genes with reported cell type information (in our curated list, see methods) are shown (Fig 3G, comprehensive figure in Fig S3).

Sensory and sensory-processing neurons in retina development are affected in Zebrafish lacking Stim2

The lincRNAs – *CR786577.1* and *CT027638.1*, and the protein-coding genes – *zgc:158463*, *arginine/serine-rich protein 1 (rsrp1)*, *potassium large conductance calcium-activated channel, subfamily M, alpha member 1a (kcnma1a)*, and *CT583728.24* were significantly upregulated (qvalue < 0.0001) in subcluster 63 (compared to the full cluster), but downregulated in subcluster 4 (Fig 3G, Fig S3, Fig 4A, Supplementary Table 3, 4 and 5). Among these genes, *kcnma1a* is known to be expressed in otic vesicle and acts upstream of or within response to auditory stimulus. Human ortholog(s) of *kcnma1a* is associated with Alzheimer's disease and idiopathic generalized epilepsy.

Among the genes significantly downregulated in the high-likelihood subcluster of cells (Fig 3G, Fig S3, Fig 4B and 4C), *MARCKS-like 1a and 1b* (*marcks1a* and *marcks1b*) are involved in neural retina development, and *myristoylated alanine-rich protein kinase C substrate b* (*marcksb*) regulates retina layer formation. Knockdown of *marcksb* has been reported to cause defects such as folded retina and larger hindbrain ventricle, whereas smaller eyes and brain have been observed in *marcks1a* and *marcks1b* knockdown. Moreover, perturbation of these genes affects neural tube closure and causes neural embryonic defects such as neuronal ectopia, failure in formation of corpus callosum, abnormality in laminations of cortex and/or retina in mice. Neuronal ectopia was also observed in the retina of *marcks1b* morphant zebrafish^{34,35}. *calbindin 2a* (*calb2a*) – reported to be highly expressed in retina and cranial ganglion, tegmentum, midbrain-hindbrain boundary, hindbrain, and spinal cord³⁶, as well as in specific neuronal cell subtypes (Fig 3G and Fig S3) – was also found to have decreased expression. *midkine b* (*mdkb*) is expressed at neural plate borders, during differentiation of retinal neurons such as ganglion cells and amacrine cells. Knockdown of *mdkb* results in drastically reduced numbers of neural crest cells and sensory neurons^{37–40}. Although the pan-neuronal marker *ELAV like neuron-specific RNA binding protein 3* (*elavl3*) was expressed in the high-likelihood subcluster, decreased expression of *ELAV like neuron-specific RNA binding protein 4* (*elavl4*), which is expressed in the nervous system and trigeminal placode (a type of neurogenic placode which promotes generation of neurons, such as trigeminal neurons, and sensory nervous system in general) was observed. *elavl4* is also a known amacrine/ganglion cell marker in zebrafish⁴¹. Among the other genes that were significantly downregulated in the high-likelihood subcluster, the following ones are also known to be expressed in the trigeminal placode: *solute carrier family 17 member 6b* (*slc17a6b*), *stathmin 1b* (*stmn1b*), *stathmin 2a* (*stmn2a*), *transmembrane protein 59-like* (*tmem59l*), *reticulon 1a* (*rtn1a*), *glycoprotein M6Ab* (*gpm6ab*), *adenylate cyclase activating polypeptide 1b* (*adcyap1b*), “aldolase C, fructose-bisphosphate, b” (*aldocb*), etc.

Among these genes, *slc17a6b* is a glutamatergic neuronal marker⁴². *tmem59l* is involved in the extension of peripheral axons in primary sensory neurons⁴³. While both of the prothymosin genes, *prothymosin alpha a* (*ptmaa*) and *prothymosin alpha b* (*ptmab*), are expressed in amacrine cells, only the *ptmaa* transcript is found in trigeminal ganglion and only *ptmab* in horizontal cells⁴⁴. A potential role of *stmn1b* in regulating neuronal differentiation by complementing the effect of *stmn4* has been reported⁴⁵. Human reticulon *RTN1* (zebrafish ortholog *rtn1a*, Fig 4C) is generally expressed in neuronal cells, shows high expression in the dendrites of purkinje cells, and the protein is enriched in the amyloid plaques in human brains with Alzheimer’s disease⁴⁶. Reticulon proteins, in general, reduce ER stress⁴⁷, are associated with ER-phagy, and defects in the process may lead to neurodegeneration⁴⁸. Involved in neuron projection development and regulation of filopodium assembly, *gpm6ab* is a Schizophrenia-associated gene⁴⁹. *Synaptosome associated protein 25a* (*snap25a*) acts as a key component in modulation of primary motor neuron development and larval locomotor activities. *adcyap1b* encodes a pituitary adenylate cyclase-activating polypeptide which functions in ischemic conditions as a neuroprotective factor^{50,51}, in retinal protection^{52,53}, immunomodulation^{54,55}, stress response⁵⁶, and others.

proprotein convertase subtilisin/kexin type 1 inhibitor, like (*pcsk1nl*) encodes a neuropeptide precursor and is expressed in adenohypophyseal placode, and cranial nerve CN-IX and CN-X. Adenohypophyseal placode consists of a specialised group of cells that gives rise to the anterior lobe of the pituitary gland and subsequently to endocrine secretory cells⁵⁷. *si:dkey-35i13.1* is expressed in Purkinje cells, granule cells, optic tectum (from ZFIN⁵⁸); the optic tectum is the primary visual center in fishes and inputs from retina and other sensory modalities are mapped over tectum. The integrated sensory information in the optic tectum undergoes visuomotor

transformation and the generated space-code is passed onto the downstream premotor assemblies to direct movement⁵⁹. Proper circuit integration of newborn neurons in the optic tectum is crucial for survival⁶⁰.

The genes significantly downregulated in the high-likelihood subcluster (subcluster 63) (some of the potential marker genes with the strongest effects discussed above) showed lower average expression than the low-likelihood subcluster (subcluster 4), and when compared alongside the expression distributions in the full cluster (Figure 4B, 4C; bottom panels), suggested absence of the cells with higher expression of those genes. We provide a categorical description of all the downregulated genes in the high-likelihood subcluster in the following section.

Ribosomal protein coding genes linked with abnormality of brain development, genes involved in oxidative phosphorylation and autophagy are downregulated in the most affected sub-cluster of cells due to Stim2 deletion

Gene set enrichment analysis revealed that the functional terms - “ribosome”, “SRP-dependent cotranslational protein targeting to membrane”, “autophagy” (member genes: *gabap12*, *gabap12*, *gabap12*, *atg13*, *fundc1*, *zgc:92606*, *gabap12*, *map1lc3b*), “autophagy -other” (*gabap12*, *gabap12*, *atg13*, *zgc:92606*, *gabap12*, *ppp2cb*), “macroautophagy” (*zgc:65894*, *dyn112b*, *gabap12*, *gabap12*, *atg13*, *fundc1*, *zgc:92606*, *vdac1*, *ube2v1*, *gabap12*, *tuba1c*, *csnk2b*, *rps27a*, *map1lc3b*, *tomm20b*), “mitophagy” (*fundc1*, *vdac1*, *csnk2b*, *rps27a*, *map1lc3b*, *tomm20b*), “mitophagy - animal” (*foxo3a*, *gabap12*, *gabap12*, *atf4b*, *fundc1*, *zgc:92606*, *uba52*, *gabap12*, *csnk2b*, *rps27a*), “oxidative phosphorylation”(*atp5pb*, *atp5l*, *atp6v1ba*, *uqcrb*, *atp6v0ca*, *atp6v1d*, *atp6v1c1a*, *atp6v1g1*, *ndufb7*, *atp6v1aa*, *atp6v1e1b*, *atp6v0cb*, *atp5mf*, *ndufb9*, *atp6ap1a*, *atp6v1b2*, *atp6v0e2*, *atp5meh*, *atp6v0d1*, *ndufa5*, *atp6v1h*, *ndufa3*, *ndufa12*, *uqcrh*, *atp6v0e1*), “Proton-transporting ATPase activity, rotational mechanism”, “Vacuolar proton-transporting V-type ATPase complex”, “phagosome”, “ROS and RNS production in phagocytes”, “insulin receptor recycling”, “iron uptake and transport”, etc were highly significantly (FDR < 0.001) enriched in genes downregulated (qvalue < 0.0001, log2 fold change < 0, 443 genes) in the high-likelihood sub-cluster of cells (Supplementary Table 6).

Significantly decreased expression of 75 ribosomal protein coding genes were detected (term – ribosome). Interestingly, other significantly enriched terms associated with abnormality in zebrafish optic tectum, retina, retinal pigmented epithelium, neurogenic placode, otic placode, nerve quality, hindbrain (from Zebrafish Phenotype Ontology terms) etc were also enriched in many of these ribosomal genes and other member genes (selected terms shown in Fig S4) . Knockdown of ribosomal protein coding genes were previously reported to generate specific phenotypes, for each gene, related to abnormal development of brain, eyes, ears and body trunk with varying degrees of effect in ~25hpf zebrafish⁶¹.

The following mitochondria-associated terms were also significantly enriched - Inner mitochondrial membrane protein complex (*atp5pb*, *atp5l*, *uqcrb*, *ndufb7*, *romo1*, *ndufb9*, *atp5meh*, *ndufa5*, *ndufa3*, *ndufa12*, *uqcrh*, *zgc:193541*), Mitochondrial protein-containing complex (*atp5pb*, *atp5l*, *uqcrb*, *ndufb7*, *romo1*, *ndufb9*, *vdac1*, *atp5meh*, *ndufa5*, *slc25a5*, *ndufa3*, *ndufa12*, *uqcrh*, *tomm20b*, *zgc:193541*), Mitochondrial proton-transporting ATP synthase complex (*atp5pb*, *atp5l*, *atp5meh*, *zgc:193541*), Mitochondrial respirasome (*uqcrb*, *ndufb7*, *ndufb9*, *ndufa5*, *ndufa3*, *ndufa12*, *uqcrh*), Mitochondrial respiratory chain complex I

(*ndufb7*, *ndufb9*, *ndufa5*, *ndufa3*, *ndufa12*), “Mitochondrial proton-transporting ATP synthase complex, coupling factor F(o)” (*atp5pb*, *atp5l*, *atp5meb*), Activation of BAD and translocation to mitochondria (*ywhah*, *ywhaqb*, *ywhae1*, *ywhaq*, *ywhag2*), Mitochondrial proton-transporting ATP synthase complex (*atp5pb*, *atp5l*, *atp5meb*, *zgc:193541*). Understandably, the member genes of these terms largely overlap with those in oxidative phosphorylation. This suggests that the mitochondrial electron transport chain, and the process of mitochondrial respiration or oxidative phosphorylation in general, is dysregulated in the neuronal cell types most affected by Stim2 deletion in (*stim2a;stim2b*)^{-/-} zebrafish.

Among other terms, “Retrograde neurotrophin signalling” (FDR=0.019) acts in development and maintenance of correct neural connectivity, and defects in this pathway are involved in many neurodegenerative disorders, such as Alzheimer's disease, Huntington's disease and psychiatric disorders, etc^{62,63}. Subsets of the genes associated with “oxidative phosphorylation” and proton-transporting ATPase activity terms were associated with “Cellular response to increased oxygen levels” (FDR=0.0034; *atp6v1g1*, *atp6v1aa*, *atp6ap1a*, *atp6v0d1*) and abnormality in retinal pigmented epithelium (FDR=0.0012; *atp6v0ca*, *atp6v1e1b*, *atp6v0d1*, *atp6v1h*). “Regulation of inflammatory response to wounding” (strength=1.68, FDR=0.0015; *mdkb*, *tmsb*, *tmsb2*, *tmsb4x*) could indicate dysregulated neuro-immune processes in the double-knockout fish.

Conclusion

We recently showed that the lack of Stim2 affects vision (with visual-motor response test and dark-light preference test) and capability for survival in low oxygen conditions in (*stim2a;stim2b*)^{-/-} zebrafish larvae³¹. Downregulation of light perception related genes, which are actively expressed in the eyes, were supported by transcriptomic data (bulk RNA-seq). Abnormality in retinal pigmented epithelium was detected with periodic acid-Schiff staining of retina sections of (*stim2a;stim2b*)^{-/-} adult zebrafish. Also, the double mutant fish had a thinner layer of ganglion cells.

In the current study, we investigated the effect of Stim2 deletion in the nervous system development of zebrafish at single cell resolution (neuronal origin cells). We identified the cell cluster with the highest mean (*stim2a;stim2b*)^{-/-} likelihood (Fig 1), and identified the cluster to be an admixture of cell types such as amacrine cells, retinal ganglion cells, neural crest cells, secondary motor neurons, and other sensory and receptor cells, based on the marker gene expression (Fig 2). The KEGG pathway - neuroactive ligand-receptor interaction - was significantly enriched in the top differentially expressed genes in the cluster. The cell cluster was then divided into three subclusters, namely - low-, mid-, and high-(*stim2a;stim2b*)^{-/-} likelihood subclusters. Comparison of the DE genes of these three subclusters revealed key genes and pathways that are perturbed in the (*stim2a;stim2b*)^{-/-} zebrafish and provide potential molecular underpinnings of the observed phenotypes.

Our results suggest that retina development, and more broadly the development of sensory and sensory processing neurons, was the most affected process in (*stim2a;stim2b*)^{-/-} zebrafish. Among the significantly downregulated genes in the high-likelihood subcluster of cells, the roles of the genes such as *marcksl1a*, *marcksl1b*, *marcksb*, *calb2a*, *mdkb*, *elavl4*, *slc17a6b*, *tmem59l*, *ptmaa*, *ptmab*, *stmn1b*, *rtn1a*, *snap25a*, *adcyp1b*, *pcsk1nl*, and *si:dkey-35i13.1* have been discussed above. The decreased expression of these genes and often unimodal distribution of expression in (*stim2a;stim2b*)^{-/-} cells compared to the higher expressions and bimodal distributions in (*stim2a;stim2b*)^{+/+} cells (Fig 4), potentially suggests reduced number of the cell types designated by the corresponding marker genes in the (*stim2a;stim2b*)^{-/-} zebrafish. We see

this pattern for the marker genes of retinal ganglion cells, cranial/trigeminal ganglion cells, amacrine cells, neural crest cells, sensory neurons and others. This provides a potential explanation of the thinner ganglion cell layer in the *(stim2a;stim2b)*^{-/-} zebrafish. Other genes that regulate development of retina, primary sensory neurons, optic tectum, neuronal differentiation, primary motor neurons and locomotor activities, endocrine secretory cells, reduction of ER stress, and neuroprotection in ischemic conditions also showed this pattern - suggesting these processes to be impaired in *(stim2a;stim2b)*^{-/-} zebrafish.

Abnormality in zebrafish optic tectum, retina, retinal pigmented epithelium, neurogenic placode, otic placode, nerve quality etc (Fig S4) and potential involvement of the ribosomal and other genes was discussed above. Large-scale downregulation of ribosomal protein coding genes in these cells also may play crucial roles in defects in the sensory nervous system development in the absence of Stim2. Neural crest cells and cranial placode cells interact with each other at the neural plate border in cranial sensory development^{64,65}. The molecular and environmental signals during such interactions orchestrate specification of vertebrate sensory regions, and development of the peripheral sensory organs in general. We observed the key genes in both of these cell types to be downregulated in the high *(stim2a;stim2b)*^{-/-} likelihood subcluster, raising the possibility of the sensory system to be affected. Moreover, a compromised optic tectum would result in impaired visuo-sensory-motor signal integration and subsequent locomotor activities - as observed in the earlier behavioural tests³¹.

Autophagy is important for maintenance of cellular homeostasis. Dysfunctional autophagy has been identified as one of the hallmarks of neurodegenerative diseases. Decreased autophagy in ageing leads to accumulation of cellular materials such as misfolded proteins and dysfunctional mitochondria, and subsequent neurodegenerative disorders such Alzheimer's disease, Parkinson's disease and others^{66,67}. Moreover, reduced oxidative phosphorylation in these cells provides insights into plausible mechanisms behind the lack of survival capability of these fish in hypoxic conditions. Abnormality in autophagy/mitophagy/ER-phagy processes could potentially lead to defects in cellular damage control. Whether downregulation of retrograde neurotrophin signalling pathway genes contributes to anxiety-like behaviour in *(stim2a;stim2b)*^{-/-} fish needs further investigation.

While not as strongly as the vision circuitry, but the olfactory granule cells (*tph2*), auditory receptor cells (*atf5b*) were also affected. The significant upregulation of the auditory circuitry gene *kcnma1a* in the high-likelihood subcluster of the *(stim2a;stim2b)*^{-/-} zebrafish may suggest presence of strengthened auditory senses in the double-knockout fish in the absence of effective vision circuitry.

Altogether, our results identify the neuronal origin cells in *(stim2a;stim2b)*^{-/-} zebrafish that are most affected due to the gene deletion. The affected cell types, key genes and processes provide valuable insights into the molecular processes behind the observed phenotypes in *(stim2a;stim2b)*^{-/-} zebrafish. Simultaneously, this study elucidates the importance of Stim2 proteins in not only regulating Ca²⁺ homeostasis, but also its complex interplay with other processes in nervous system development and its potential therapeutic value in neurodegenerative diseases in humans.

Materials and methods

scRNA-seq data analysis

Identification of cell clusters

We obtained the primary assembly of zebrafish genome fasta (Danio rerio GRCz11) and transcriptome annotation files from ensembl (release-105), filtered the transcriptome to keep only protein-coding and lincRNA genes, and prepared the reference with 10x Genomics Cell Ranger 6.1.2⁶⁸. Raw reads from fastq files were mapped to the aforementioned zebrafish reference genome and gene expression were quantified with Cell Ranger count, for each sample. We then analysed the count matrices with scanpy⁶⁹ and scvi-tools⁷⁰. Potential cell doublets were removed with Solo⁷¹. In the preprocessing step, we then integrated all datasets from eight samples (four replicates each for transgenic/control and stim2-knock out conditions), filtered out genes that were not expressed in at least three cells, and kept only the cells that expressed at least 200 genes and contained less than 10% mitochondrial genes. The count data were normalised, pseudocount (of 1) added, and log-transformed. Highly variable genes were computed with scanpy with `n_top_genes=3000` and `flavor= 'seurat_v3'`. We then set up and trained a single cell variational inference (scVI) model⁷², and extracted the latent representation of each cell from our data which was subsequently used (with parameter “use_rep”) for computing the neighbourhood graph with scanpy. The neighbourhood graph was embedded using UMAP and Leiden clustering (resolution = 1) was performed. Scvi-tools was used for identification of marker genes per cluster, and differential gene expression in general (between (*stim2a;stim2b*)^{-/-} and control for each cluster, as well as for all clusters/ the entire dataset).

We utilised the curated gene expression data from the zebrafish information network (ZFIN) database⁵⁸ for all cell types in the zebrafish brain, and compiled a list of marker genes with the highly expressed genes in each cell type at 5 dpf developmental stage. In the absence of data of highly expressed genes at the 5dpf stage, we utilized the highly expressed genes at the nearest time-point available. Furthermore, cell type information for the marker genes in our dataset were extracted from scientific literature and added to the list mentioned above (Supplementary Table ***).

Quantifying the effects of stim2 knockout at single cell resolution

We used the preprocessed, normalised and logarithmized data layer from the integrated data (step above) as the input data, and the annotation of observations layer as metadata for this step. PCA was calculated and lower dimensional embedding of the data was performed with PHATE⁷³. We then utilized the MELD algorithm³³ for quantifying the effects of the gene knockout. We followed the steps recommended by the authors of the algorithm. A search for optimal filter and graph parameters specific for our dataset for the MELD algorithm was performed. The best set of parameter values were: `beta = 42.0`, `knn = 20.0`, `MSE = 0.002388`, and `seed = 12.0`. We subsequently used these values to build the graph for MELD and estimate densities for each sample. The densities were compared among the samples, and L1 normalisation was applied across the densities for each sample in order to compute the sample associated relative likelihoods for each condition. We finally used the average likelihood of the four (*stim2a;stim2b*)^{-/-} samples (replicates) as the measure of the perturbation.

Subclustering of clusters of interest was performed with vertex frequency clustering⁷⁴ (VFC), denoising and imputation of the cell count matrix was done with MAGIC⁷⁵, and differential gene expression at the subcluster level was performed with diffxpy (<https://github.com/theislab/diffxpy>). Gene set enrichment analysis was performed on STRING database⁷⁶ and g:Profiler⁷⁷.

Figure legends

Figure 1: Identification of the UMAP cluster with highest $(stim2a;stim2b)^{-/-}$ likelihood. A) Schematic showing the cells preparation and processing of the 5dpf zebrafish brain, FACS sorting, library preparation and data analysis. B) The cell neighbourhood graph from our single cell transcriptomic data in UMAP embedding. Leiden clusters (resolution=1) are indicated with cluster numbers and corresponding colors. C) PHATE embedding of the data. Neuronal origin cells from *Tg* and $(stim2a;stim2b)^{-/-}$ zebrafish are coloured in blue (*Tg*) and orange (*stim2ab*). D) Mean (left) and standard deviation (right) of $(stim2a;stim2b)^{-/-}$ likelihood of the cells are shown on PHATE embedding. E) Mean $(stim2a;stim2b)^{-/-}$ likelihood of cells from each Leiden cluster. Clusters sorted by increasing mean likelihood.

Figure 2: Marker genes of the cell cluster with highest mean $(stim2a;stim2b)^{-/-}$ likelihood and cell type(s) annotation. A) Expression (tracks plot) of the marker genes of UMAP (Leiden) cluster 5 across all clusters. B) Word Cloud of cell type information of the top marker genes of UMAP (Leiden) cluster 5.

Figure 3: Identification of the subcluster of cells most affected by *Stim2* deletion. A) Visualization of $(stim2a;stim2b)^{-/-}$ likelihood of all cells from each of the UMAP clusters on PHATE embedding. Please note that the cluster numbers here reflect the sorting performed in Fig2D. Here, Cluster 29 is the UMAP Cluster 5 from Fig-2. B) The $(stim2a;stim2b)^{-/-}$ likelihood scores are shown for UMAP Cluster 5 (Cluster 29 in Fig 3A). C) Different numbers of subclusters of the UMAP Cluster 5 with vertex frequency clustering. Three subclusters distinguished the cells according to $(stim2a;stim2b)^{-/-}$ likelihood (marked in red). D) The $(stim2a;stim2b)^{-/-}$ likelihood of cells from all the subclusters (at optimal level of grouping) from each of the selected 14 UMAP clusters (based on the number of member cells with high likelihood). The subclusters were sorted by mean $(stim2a;stim2b)^{-/-}$ likelihood of the subclusters. The three subclusters from UMAP Cluster 5: 4, 44, and 63 – represent the low-, mid-, and high-likelihood subclusters, respectively. E) The same subclusters, colored in blue, orange and green, respectively, are shown on PHATE embedding. F) Scatterplot of individual $(stim2a;stim2b)^{-/-}$ likelihood of cells from each subcluster. Cells are colored based on their sample of origin. TgX: *Tg* samples, stim2abX: $(stim2a;stim2b)^{-/-}$ samples. G) Marker plot of subclusters with low, mid, and high $(stim2a;stim2b)^{-/-}$ likelihood. Differentially expressed (DE) genes (qvalue < 0.0001; gene name on y-axis, left) with cell type information (y-axis, right) available in our curated list. Color of each point indicates the expression of each gene in each cluster (darker and brighter colors indicate lower and higher expression, respectively). The size of each point indicates how differentially expressed each gene is in each cluster. Subcluster 4, 44, and 63 are the low, mid, and high $(stim2a;stim2b)^{-/-}$ likelihood subclusters, respectively.

Figure 4: Expression changes between conditions in the full cluster vs in VFC clusters. A) Genes with differentially increased expression in subcluster 63. B) and C) Selected top genes with differentially decreased expression in subcluster 63 (except *elavl3*, which is included for comparison with *elavl4*). In each subpanel, the rows (top to bottom) show the expression distribution of the corresponding genes in subcluster 4, 44, and 63, and in the full cluster (UMAP

Leiden cluster 5), respectively. Subclusters 4, 44, and 63 are the low, mid, and high (*stim2a;stim2b*)^{-/-} likelihood subclusters, respectively. Distribution and mean line of gene expression in cells from control (*Tg*) and (*stim2a;stim2b*)^{-/-} Zebrafish are shown in blue and red, respectively.

Supplementary Figure S1: MELD Results. A) Sample associated density estimates computed with MELD algorithm are visualized on PHATE plot. B) Sample associated relative likelihood estimates across replicates of (*stim2a;stim2b*)^{-/-} Zebrafish shown on PHATE plot.

Supplementary Figure S2: VFC clustering Results. VFC clustering of all Leiden clusters at 2-5 levels of sub-clustering are shown. The chosen level of sub-clustering for each cluster is outlined with a red box.

Supplementary Figure S3: Marker plot of subclusters with low, mid, and high (*stim2a;stim2b*)^{-/-} likelihood. Differentially expressed (DE) genes (qvalue < 0.0001; y-axis, left) with cell type information (y-axis, right), as well as the non-redundant set of top 50 (by log fold change) up- and down-regulated DE genes for each cluster are shown. Genes without known cell type information (in our curated list) are marked as “unknown” (y-axis, right). Color of each point indicates the expression of each gene in each cluster (darker and brighter colors indicate lower and higher expression, respectively). The size of each point indicates how differentially expressed each gene is in each cluster. Subclusters 4, 44, and 63 are the low, mid, and high (*stim2a;stim2b*)^{-/-} likelihood subclusters, respectively.

Supplementary Figure S4: Phenotype ontology terms and member genes. Selected Zebrafish Phenotype Ontology terms significantly enriched in significantly downregulated genes in the high (*stim2a;stim2b*)^{-/-} likelihood subcluster (subcluster 63) are shown along with their member genes in this binary clustermap. If a gene is a member of the ontology term, the membership is marked in black.

References

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